

Laboratory Revenue and Operations Dashboard

Healthcare Analytics Project Report

Prepared by:

Maryam Abubakar

Medical Laboratory Scientist | Healthcare Data Analyst

Project Type:

Healthcare Analytics Dashboard

Tools Used:

Microsoft Excel | PivotTables | Data Visualization

Project Period:

January 2025 – December 2025

Date Created:

July 2026

Project Report

1. Project Overview

This project presents an interactive Microsoft excel healthcare analytics dashboard developed to analyse laboratory revenue performance, test demand patterns, branch performance, departmental contribution, and customer distribution.

The dashboard was created using a synthetic laboratory sales dataset to demonstrate healthcare data analysis and visualisation skills.

2. Project Objectives

The objectives of this project were to:

- Analyse overall laboratory revenue performance.
- Identify the most requested tests.
- Evaluate revenue contribution by laboratory branch.
- Compare revenue performance across laboratory departments.
- Understand customer distribution patterns.
- Generate actionable insights to support operational decision-making

3. Dataset Description

Dataset Name: Laboratory Sales Dataset

Datatype: Synthetic healthcare sales data

Records: 1,250 laboratory transactions

Key Variables:

Column	Description
Transaction ID	Unique transaction identifier
Test Date	Date test was performed
Branch	Laboratory branch location
Department	Laboratory department
Test Name	Laboratory investigation performed
Patient type	Customer category
Payment Method	Mode of payment
Revenue	Income generated from test

4. Tools Used

Microsoft Excel

- Data cleaning
- Pivot Tables
- Pivot Charts
- Dashboard design
- Data visualization

Analytical Techniques

- KPI calculation
- Trend analysis
- Revenue comparison
- Customer segmentation

5. Dashboard Features

The dashboard contains:

Key performance indicators (KPIs)

- Total revenue
- Total tests performed
- Average revenue per month
- Average revenue per test
- Average tests per month
- Most requested test
- Highest revenue branch
- Revenue growth

Visual Analysis

Monthly Revenue Trend

- Shows revenue performance across January - December 2025

Top 10 Requested Tests

- Identifies laboratory investigations with the highest patient demand

Branch Revenue Share

- Compares revenue contribution between laboratory branches

Revenue by Department

- Shows departmental financial performance

Customer Distribution

- Displays patient source distribution:
 - Walk-in
 - Corporate
 - Referral

6. Key Findings

Revenue Performance

- Total laboratory revenue generated was **~~N~~ 5.60M**.
- Average revenue per test was **~~N~~4,483**.
- Revenue increased by **3.3%** compared with the previous quarter

Branch Performance

- City branch generated the highest revenue contribution (35.6%).
- City branch showed strong operational performance compared with other branches.

Test Demand

- Malaria test was the most requested investigation with **113 requests**.
- Other high-demand tests included:
 - Hepatitis B Test
 - Stool Culture
 - Kidney Function Test

Department Performance

- **Clinical Chemistry** generated the highest departmental revenue (**~~N~~2.25M**)

Customer Analysis

- Customer distribution was balanced among:
 - Walk-in patients
 - Corporate clients
 - Referral patients

7. Recommendations

Based on dashboard insights:

1. Maintain adequate reagent availability for high-demand tests such as malaria testing.
2. Prioritise resources allocation to high performing departments such as clinical chemistry.
3. Investigate strategies to improve revenue contribution from lower-performing branches.
4. Strengthen corporate and referral partnerships to increase patient volume.
5. Continue monitoring monthly revenue trends for operational planning.

8. Conclusion

The dashboard demonstrates how healthcare data analytics can transform laboratory operational data into meaningful business and clinical insights

The project highlights the importance of data driven decision-making in improving laboratory efficiency, resource allocation, and service delivery

9. Author

Maryam Abubakar

Medical Laboratory Scientist (BMLS, UDUS)

Aspiring Healthcare Data Analyst